Some Computational QM Principles - Project 2

CURIE SP26 - Programming in Computational Quantum Chemistry

The first two sections will be based on an existing code from jupyter notebook cells. (You may run these as a single python script as well) The last section is for setting up the basis functions for the long awaited Hartree-Fock code.

1 Variational Principle

1.1 1D Harmonic (Background)

The 1D Quantum Harmonic Oscillator:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\frac{1}{2}m\omega^2x^2\right)\psi(x)=E\psi(x)$$

has the energy and wavefunctions:

$$E=\hbar\omega\left(n+\frac{1}{2}\right) \qquad \psi_n(x)=\frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-\frac{m\omega x^2}{2\hbar}}H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)$$

Where $H_n(x)$ is the Hermite polynomial. To simplify the problem we remove all constants and units by defining $x^*=x/l$ and $\hat{H}^*=\hat{H}/\epsilon$ where $\epsilon=\hbar^2/ml^2=m\omega^2l^2$. The reduced Hamiltonian and its solutions are:

$$\hat{H}^*=-\frac{1}{2}\frac{d^2}{dx^2}+\frac{1}{2}(x^*)^2$$

$$E_n^*=n+\frac{1}{2} \qquad \psi_n(x^*)=\frac{1}{\sqrt{2^n n!}}\pi^{-1/4}e^{-\frac{(x^*)^2}{2}}H_n(x^*)$$

with the first three energies and wavefunctions shown in Table 1.1.

n	E_n^*	$\psi_n(x^*)$
0	$\frac{1}{2}$	$\pi^{-1/4} e^{-\frac{(x^*)^2}{2}}$
1	$\frac{3}{2}$	$\pi^{-1/4}\sqrt{2}\, x^*\, e^{-\frac{(x^*)^2}{2}}$
2	$\frac{5}{2}$	$\pi^{-1/4}\frac{1}{\sqrt{2}}\left(2(x^*)^2 - 1\right) e^{-\frac{(x^*)^2}{2}}$

Table 1: First three energy and wavefunctions of the harmonic oscillator in reduced units.

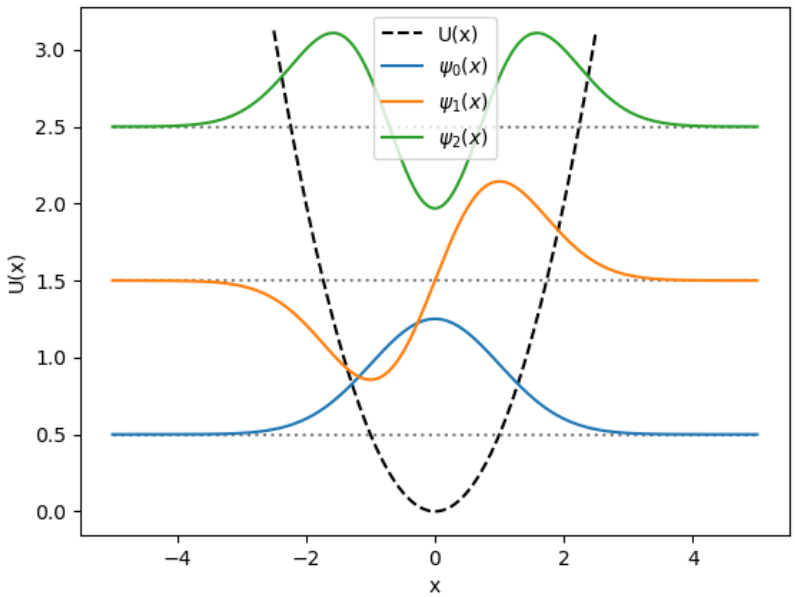

Figure 1: Plot of the first three harmonic oscillator wavefunctions.

1.2 Approximating with the Linear Variational Principle

I recommend reading Section 4.1 on the variational principle. We will approximate the wavefunction and solution of the first three wavefunctions of the Harmonic oscillators with a linear combination of gaussian functions serving as our basis of "orbitals":

$$\psi(x) = \sum_{\mu}^{n} c_\mu \phi_\mu(x) \quad \text{where} \quad \phi_\mu(x) = \exp[-\alpha(x - x_a)^2] \qquad \alpha = 2$$

The gaussian functions are evenly spaced and offset from the center x by x_A with a fixed width parameter $\alpha = 2$. We will use a different number of basis functions ($n = 5, 10, 15, 20$) and compare their accuracy in approximating the energy and wavefunctions.

For computing the matrix elements/integrals, the analytical solutions are given as:

$$
\begin{aligned}
S_{\mu\nu} = \langle \phi_\mu | \phi_\nu \rangle &= \int_{-\infty}^{+\infty} dx \exp\left(-\alpha(x - x_A)^2\right) \exp\left(-\alpha(x - x_B)^2\right) \\
&= \sqrt{\frac{\pi}{2\alpha}} \exp\left(-\frac{\alpha}{2}(x_A - x_B)^2\right) \\
H_{\mu\nu} = \langle \phi_\mu | H | \phi_\nu \rangle &= \int_{-\infty}^{+\infty} dx \exp\left(-\alpha(x - x_A)^2\right) \left(-\frac{d^2}{dx^2} + \frac{1}{2}x^2\right) \exp\left(-\alpha(x - x_B)^2\right) \\
&= \frac{1}{2} S_{\mu\nu} \left[\alpha - \alpha^2 (x_A - x_B)^2 + \frac{1}{4}(\frac{1}{2} + (x_A + x_B)^2)\right] \\
&= \frac{1}{2} \left[\sqrt{\frac{\pi}{2\alpha}} \exp\left(-\frac{\alpha}{2}(x_A - x_B)^2\right)\right] \left[\alpha - \alpha^2 (x_A - x_B)^2 + \frac{1}{4}(\frac{1}{2} + (x_A + x_B)^2)\right]
\end{aligned}
$$

with the overlap and the Hamiltonian matrices formulated as:

$$
\mathbf{S} = \begin{bmatrix} \langle \phi_1 | \phi_1 \rangle & \cdots & \langle \phi_1 | \phi_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \phi_n | \phi_1 \rangle & \cdots & \langle \phi_n | \phi_n \rangle \end{bmatrix} \qquad \mathbf{H} = \begin{bmatrix} \langle \phi_1 | \hat{H} | \phi_1 \rangle & \cdots & \langle \phi_1 | \hat{H} | \phi_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \phi_n | \hat{H} | \phi_1 \rangle & \cdots & \langle \phi_n | \hat{H} | \phi_n \rangle \end{bmatrix}
$$

Here are some more explicit instructions:

- Implement code that evaluates $S_{\mu\nu}$ and $H_{\mu\nu}$ given the analytical solution of the integrals above and build $\mathbf{S}$ and $\mathbf{H}$ with `n_basis = [5, 10, 15, 20]` gaussian basis functions evenly by x_A in the range $[-5, 5]$ and a fixed width parameters $\alpha = 2$.
- Build the orthogonalizing transformation $\mathbf{X}$ with the $\mathbf{S}$ you built (read Section 4.2) and orthogonalize your Hamiltonian $\mathbf{H}' = \mathbf{X}^\mathbf{T}\mathbf{H}\mathbf{X}$.
- Evaluate the energy and variationally optimized coefficients by diagonalizing $\mathbf{H}'$ (remember the transform the coefficients back into the original basis $\mathbf{c} = \mathbf{X}\mathbf{c}'$). Compare the accuracy of the energy of the first three states.
- Build the first three approximate HO wavefunctions by taking linear combinations of the variationally optimized coefficients and its corresponding gaussian basis functions and plot it over the original exact wavefunctions. Repeat for each `n_basis` and compare their accuracy. (The resulting function may be flipped randomly, just plot the inverse of the function in that case)

2 Correlation

We will now add a alight complication to our

3 An Introduction to The Project

The MO theory, particularly LCAO-MO (Linear Combination of Atomic Orbitals - Molecular Orbital), is a general idea of constructing molecules from atoms. More specifically, coefficients of a linear combinations of atomic orbitals $(1s, 2s, 2p, ...)$ are used to construct molecular orbitals $(\sigma_{1s}, \sigma^*_{1s}, \sigma_{2s}, ...)$, where orbitals are functions (eigenfunctions of the Hamiltonian of the atom/molecule system) which describe the electron wavefunction in atoms and molecules. With such procedure, one could start with just a list of atoms and its positions and build/simulate molecules, often uncovering insights not prominent in experimental work.

Hartree-Fock is one specific, computationally feasible formulation to approximate the energy and wavefunction of the the many-electron molecule within the broad idea of MO theory. This is an ab-initio method, meaning that it is grounded in entirely in theory rather than empirical observations or fitting. A defining feature of the Hartree-Fock method is that it is a mean-field or a self-consistent field theory, similar to the correlated HO system in Section 2, where the many-electron problem is simplified by averaging the interaction of all other electrons into a "mean-field" when solving the Hamiltonian for one of the electrons. This self-consistent field (SCF) procedure is a major component of HF, apart from the basis set construction and matrix element/integral computation.

First, in order to construct molecular orbitals (which you might guess involves a variational minimization of a some Hamiltonian which contains correlated operators), a basis of atomic orbitals are needed. Previously

4 Additional notes

4.1 Variational Principle

Although analytically solving the Time-Independent Schrodinger Equation gives us the exact formula for the energy and wavefunctions, we often have to approximate the energy of a system with an approximate wavefunction because the analytical solution is either too difficult or impossible to evaluate. (E.g. the two electron potential $\int_{-\infty}^{\infty} dr_i dr_j 1/r_{ij}$ for electron i and j in atoms/molecules with many electrons)

The variational principle is a powerful statement/condition useful for approximating the ground state energy. The variational principle states that the trial energy eigenvalue $E[\psi_t]$ will always be greater than or equal to the true ground state energy eigenvalue E_0 of a system described by the Hamiltonian $\hat{H}$, or:

$$E[\psi] = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0$$

Visually, you can imagine that the true groundstate wavefunction is a minima of some convex energy potential well and one can achieve the best guess wavefunction eigenstate by simply minimizing the energy eigenvalue in the TISE without worrying about wandering off below the true ground state.

In practice, your Anzts (guess wavefunction) contains a single/multiple parameters that can be optimized to minimize the energy eigenvalue in the TISE. We will work exclusively with Anzts with a linear combination of basis functions:

$$|\psi\rangle = \sum_{\mu} c_\mu |\phi_i\rangle = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \begin{matrix} |\phi_1\rangle \\ |\phi_2\rangle \\ \vdots \\ |\phi_n\rangle \end{matrix} \quad \text{or} \quad \psi(x) = \sum_{\mu} c_\mu \phi_\mu(x) = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \begin{matrix} \phi_1(x) \\ \phi_2(x) \\ \vdots \\ \phi_n(x) \end{matrix}$$

where $\{\phi_1, \phi_2, ...\phi_n\}$ defines the basis set and the coefficients/vector $\{c_1, c_2, ..., c_n\}$ are the variational parameters.

Solving the TISE with this finite, coefficient vector requires transforming the Hamiltonian into a matrix via evaluating/integrating the matrix elements. This transformation can be

determined by plugging in the Anzats into the TISE:

$$\hat{H}|\psi\rangle = E|\psi\rangle \qquad |\psi\rangle = \sum_{\mu} c_\mu \phi_\mu(x) = \sum_{\mu} c_\mu |\phi_\mu\rangle$$

$$\langle\psi|\hat{H}|\psi\rangle = \langle\psi|E|\psi\rangle$$

$$\left(\sum_{\mu} c_\mu^* \langle\phi_\mu|\right) \hat{H} \left(\sum_{\nu} c_\nu |\phi_\nu\rangle\right) = E \left(\sum_{\mu} c_\mu^* \langle\phi_\mu|\right) \left(\sum_{\nu} c_\nu |\phi_\nu\rangle\right)$$

$$\sum_{\mu\nu} c_\mu^* \langle\phi_\mu|\hat{H}|\phi_\nu\rangle c_\nu = E \sum_{\mu} c_\mu^* \langle\phi_\mu|\phi_\nu\rangle c_\nu$$

$$\sum_{\mu\nu} c_\mu^* H_{\mu\nu} c_\nu = E \sum_{\mu\nu} c_\mu^* S_{\mu\nu} c_\nu$$

$$\mathbf{C}^\dagger \mathbf{H} \mathbf{C} = \mathbf{C}^\dagger \mathbf{S} \mathbf{C} \mathbf{E}$$

$$\mathbf{H}\mathbf{C} = \mathbf{S}\mathbf{C}\mathbf{E}$$

If we do not left multiply the resulting vector of $\mathbf{Hc}$ with $\mathbf{c}^\dagger$, we get the matrix vector equation $\mathbf{Hc} = E\mathbf{Sc}$ where $\mathbf{S} = \sum_{\mu\nu} S_{\mu\nu} = \sum_{\mu\nu} \langle\phi_\mu|\phi_\nu\rangle$ is the overlap matrix. The explicit matrix vector form of this equation looks like:

$$\begin{bmatrix} \langle\phi_1|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_1|\hat{H}|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_n|\hat{H}|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = E \begin{bmatrix} \langle\phi_1|\phi_1\rangle & \cdots & \langle\phi_1|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\phi_1\rangle & \cdots & \langle\phi_n|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$\begin{bmatrix} \langle\phi_1|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_1|\hat{H}|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\hat{H}|\phi_1\rangle & \cdots & \langle\phi_n|\hat{H}|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \langle\phi_1|\phi_1\rangle & \cdots & \langle\phi_1|\phi_n\rangle \\ \vdots & \ddots & \vdots \\ \langle\phi_n|\phi_1\rangle & \cdots & \langle\phi_n|\phi_n\rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \begin{bmatrix} E_1 & & \\ & \ddots & \\ & & E_n \end{bmatrix}$$

Note that although the overlap of the variationally minimized wavefunction $\langle\psi|\psi\rangle$ must by orthonormal, the basis functions themselves $\langle\phi_\mu|\phi_\nu\rangle$ don't necessarily have to be. This is why $S_{\mu\nu} = \langle\phi_\nu|\phi_\nu\rangle$ doesn't necessarily have to be orthonormal, but $\langle\psi|\psi\rangle = \sum_{\mu\nu} c_\mu^* \langle\phi_\mu|\phi_\nu\rangle c_\nu = \mathbf{c}^\dagger \mathbf{S} \mathbf{c} = 1$ must be true.

The general procedure for this linear variational principle method follows:

1. Select a trial wavefunction, for the Variational principle to work, the wavefunction

2. Compute (integrate) the matrix elements of $\mathbf{S}$ and $\mathbf{H}$ where $S_{\mu\nu} = \langle\phi_\mu|\phi_\nu\rangle$ and $H_{\mu\nu} = \langle\phi_\mu|\hat{H}|\phi_\nu\rangle$.

3. Minimize the eigenvalue E by solving the matrix vector equation $\mathbf{HC} = \mathbf{SCE}$ or $\mathbf{H'C'} = \mathbf{C'E}$ if orthogonalized.

4.2 Orthogonalization

To solve the $\mathbf{HC} = \mathbf{SCE}$ as an eigenvalue problem, one can modify the equation to $\mathbf{S}^{-1}\mathbf{HC} = \mathbf{CE}$. However, this requires a direct inversion of $\mathbf{S}^{-1}$ which increases/amplifies the numerical errors if $S_{\mu\nu}$ is small (aka the basis functions are far apart with little overlap). This direct inversion can be avoided by constructing some orthogonalizing transformation $\mathbf{X}$ such that $\mathbf{X^T SX} = \mathbf{I}$. (You can see that the overlap matrix of an orthonormal basis is the identity matrix) With such transformation, the overlpan matrix disappears and the equation is transformed into $\mathbf{H'C'} = \mathbf{C'E}$ where $\mathbf{H'} = \mathbf{X^T HX}$ and $\mathbf{C} = \mathbf{XC'}$.

One possible choice of such transformation, and a surprisingly simple choice, is $\mathbf{X} = \mathbf{S}^{-1/2}$. Plugging this into the equation above that this does indeed orthogonalize the basis vectors: $\mathbf{X^T SX} = (\mathbf{S}^{-1/2})^{\mathbf{T}}\mathbf{SS}^{-1/2} = \mathbf{S}^{-1/2}\mathbf{SS}^{-1/2} = \mathbf{S}^0 = \mathbf{I}$. Once again, to avoid the direct inversion of $\mathbf{S}$, we will transform it into the diagonal matrix $\mathbf{s}$ for the inversion and transform it back into $\mathbf{X}$.

Here is a more explicit proceedure for building and using an orthogonaling transofrmation:

1. Diagonalize $\mathbf{S}$ to find diagonalizing transformation (eigenvectors) $\mathbf{U}$ and the diagonals $\mathbf{s}$ (eigenvalues). ($\mathbf{SU} = \mathbf{Us}$)

2. Take the inverse square root of the diagonals $\mathbf{s}^{-1/2}$ and transform back into the original basis to find $\mathbf{X}$. Since $\mathbf{S} = \mathbf{UsU^T}$ (right multiply $\mathbf{U^T}$ to the eigenvalue equation $\mathbf{SU} = \mathbf{Us}$), $\mathbf{X} = \mathbf{Us}^{-1/2}\mathbf{U^T}$.

3. Transofrm the Hamiltonian into the orthogonal basis $\mathbf{H'} = \mathbf{X^T HX}$ and diagonalize the orthogonalized Hamiltonian to obtain the energies $\mathbf{E}$ and the orthogonalized coefficients $\mathbf{c'}$. ($\mathbf{H'C'} = \mathbf{C'E}$)

4. Transform the coefficient back into the original basis $\mathbf{C} = \mathbf{XC'}$ to obtain the coefficients $\mathbf{C}$ back.